

Sushi variegato (yutaka)

Yuta Takaya has to serve sushi to his guests. At this moment he has a long cutting board containing, arranged in a row, N pieces of sushi. There are different varieties of sushi: the i -th piece is of type $V[i]$.

Yuta plans to make some cuts on the cutting board with his katana (possibly 0, possibly $N - 1$) between adjacent pieces of sushi, to obtain a certain number of "tagliarini" to bring to the table: each tagliarini will contain a certain number (not zero) of adjacent sushi pieces. Yuta has decided that, to avoid guests getting bored, no two tagliarini should contain the same type of sushi.

How many different ways can Yuta Takaya divide the cutting board? Since this number can be very large, please return it modulo $10^9 + 7$ (1 000 000 007).

Implementation

You will need to submit exactly one file with the .cpp or .c extension.

👉 Among the attachments to this task you will find a template (yutaka.cpp and yutaka.c) with an implementation example.

You will need to implement the following function:

Size function

C/C++	int taglia(int N, int V[]);
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- The whole N represents the number of pieces of sushi.
- The array $V[]$, indexed from 0 to $N - 1$, contains the types of sushi. The i -th piece is of type $V[i]$.
- The function should return a single integer: the number of ways Yuta Takaya can make cuts, modulo $10^9 + 7$.

A simplified version of the grader used during evaluation is available in the directory for this problem, which you can use to test your solutions locally. The sample grader reads input data from stdin, calls the taglia() function, and writes its return value to stdout.

If you want to generate an input file, stdin must have this format:

- Line 1: contains the entire N .
- Line 2: contains the integers $V[0], V[1], \dots, V[N - 1]$, separated by a space.

stdout will have the following format:

- Line 1: contains the integer returned by size().

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Hiring

- $1 \leq N \leq 1\,000\,000$
- $1 \leq \ln[i] \leq 1\,000\,000$ per $i = 0, 1, \dots, N - 1$

Score assignment

Your program will be tested on several test cases grouped into subtasks. To get a score for a subtask, you must successfully complete all of its tests.

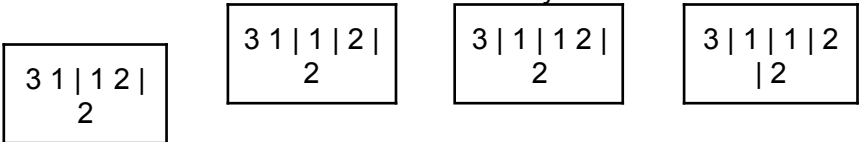
- **Subtask 1 [0 points]**: Example cases.
- **Subtask 2 [17 points]**: $N \leq 15$.
- **Subtask 3 [10 points]**: $N \leq 2,000$ and, for each type of sushi, there are at most two pieces of that type.
- **Subtask 4 [22 points]**: $N \leq 2\,000$.
- **Subtask 5 [15 points]**: $N \leq 100\,000$.
- **Subtask 6 [36 points]**: No specific limitations.

Input/output examples

5 3 1 1 2 2	4
7 1 1 1 1 1 1 1	1

Explanation

In the **first example case** There are 4 sets of cuts that satisfy the conditions:



In the **second example**, All sushi pieces are the same type, therefore there is only one way to make the cuts so that each cutter contains pieces of a different type.

